

ETHIOPIAN PAY-LATER PLATFORM

Feature-by-Feature Functional Specification

Detailed functional contract for every major product feature

Ethiopia-designed • Addis Ababa-first

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CONFIDENTIAL WORKING DOCUMENT

Document Control

Version	Date	Prepared By	Status	Summary
1.0	21 August 2026	Eleanor Tefera (Fintech)	Working baseline	Initial researched package release

Research Status Legend

VERIFIED	Directly supported by an official/primary source.
REPORTED	Supported by a reputable third-party source.
INFERENCE	Reasonable conclusion from available evidence.
ASSUMPTION	Design assumption for planning; not presented as market fact.
REQUIRES CONFIRMATION	Needs provider, bank, NBE, legal or business-partner confirmation.

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1. How to Use This Document

Each feature follows the common requirement standard requested for the project. Feature IDs are stable cross-document references. Provider-dependent features are specifications for our internal behavior; they do not assert that an external provider has approved the integration.

2. Cross-Cutting Rules

- KYC ≠ credit approval.
- First-time credit requires CRD-004 decision.
- No merchant release when required first payment is PENDING/UNKNOWN.
- Financial writes are idempotent and ledger-backed.
- Provider-specific capabilities are configuration-gated.
- Every privileged/manual override needs reason and audit evidence.
- All features expose explicit loading/pending/error/success states.

3. Detailed Feature Specifications

AUTH-001 Registration

Field	Specification
Feature ID	AUTH-001
Feature Name	Registration
Purpose	Create a secure customer account using verified contact information.
Actors	Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates registration → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.

Acceptance Criteria	Registration completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.
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AUTH-002 Login

Field	Specification
Feature ID	AUTH-002
Feature Name	Login
Purpose	Authenticate returning users with role-appropriate controls.
Actors	Customer / Merchant / Admin
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates login → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
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Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Login completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

AUTH-003 OTP

Field	Specification
Feature ID	AUTH-003
Feature Name	OTP
Purpose	Verify possession of a phone number and sensitive actions.
Actors	Customer / Merchant
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates otp → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review;

	customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	OTP completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CUS-001 Customer Profile

Field	Specification
Feature ID	CUS-001
Feature Name	Customer Profile
Purpose	Maintain identity, contact, preferences and account status.
Actors	Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates customer profile → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.

Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Customer Profile completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

KYC-001 Identity Verification

Field	Specification
Feature ID	KYC-001
Feature Name	Identity Verification
Purpose	Verify identity before Pay-Later eligibility.
Actors	Customer / Compliance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates identity verification → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Identity Verification completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

KYC-002 Fayda Verification

Field	Specification
Feature ID	KYC-002
Feature Name	Fayda Verification
Purpose	Use Fayda/accepted digital identity only if legally and technically available.
Actors	Customer / Compliance
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled

	when configured and confirmed.
Main Flow	User initiates fayda verification → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Fayda Verification completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

MER-001 Merchant Registration

Field	Specification
Feature ID	MER-001
Feature Name	Merchant Registration
Purpose	Create merchant account and capture business information.
Actors	Merchant Owner
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates merchant registration → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.

Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Merchant Registration completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

MER-002 Business Verification

Field	Specification
Feature ID	MER-002
Feature Name	Business Verification
Purpose	Verify merchant legitimacy before BNPL transactions.
Actors	Merchant / Compliance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates business verification → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Business Verification completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

MER-003 Branch Management

Field	Specification
Feature ID	MER-003
Feature Name	Branch Management

Purpose	Manage merchant branches and operating status.
Actors	Merchant Owner / Manager
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates branch management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Branch Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

MER-004 Employee Management

Field	Specification
Feature ID	MER-004
Feature Name	Employee Management
Purpose	Invite, role and deactivate merchant employees.
Actors	Merchant Owner / Manager
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates employee management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state

	transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Employee Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

MER-005 — Product Management

Field	Specification
Feature ID	MER-005
Feature Name	Product Management
Purpose	Maintain optional catalog for merchants that need it.
Actors	Merchant
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates product management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Product Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

MER-006 — Quick Sale

Field	Specification
Feature ID	MER-006
Feature Name	Quick Sale
Purpose	Create a sale from description + amount without full catalog/POS.
Actors	Merchant
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates quick sale → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Quick Sale completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

QR-001 — Merchant QR

Field	Specification
Feature ID	QR-001
Feature Name	Merchant QR
Purpose	Start a merchant-linked checkout by QR.
Actors	Merchant / Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates merchant qr → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without

	traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Merchant QR completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

QR-002 — Transaction QR

Field	Specification
Feature ID	QR-002
Feature Name	Transaction QR
Purpose	Open a specific sale safely using short-lived QR.
Actors	Merchant / Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates transaction qr → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.

Acceptance Criteria	Transaction QR completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.
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DISC-001 — Merchant Discovery

Field	Specification
Feature ID	DISC-001
Feature Name	Merchant Discovery
Purpose	Find participating merchants and categories.
Actors	Customer
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates merchant discovery → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Merchant Discovery completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CRD-001 — Customer Eligibility

Field	Specification
Feature ID	CRD-001
Feature Name	Customer Eligibility
Purpose	Determine whether Pay-Later is available responsibly.
Actors	Customer / Risk
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates customer eligibility → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only

	when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Customer Eligibility completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CRD-002 — Credit Limit

Field	Specification
Feature ID	CRD-002
Feature Name	Credit Limit
Purpose	Manage approved, available, reserved and restricted exposure.
Actors	Customer / Risk
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates credit limit → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID,

	policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Credit Limit completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CRD-003 — Transaction Decision

Field	Specification
Feature ID	CRD-003
Feature Name	Transaction Decision
Purpose	Reassess each purchase even when a customer has a limit.
Actors	Customer / Risk
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates transaction decision → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Transaction Decision completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CRD-004 — First-Time Eligibility

Field	Specification
Feature ID	CRD-004
Feature Name	First-Time Eligibility
Purpose	Approve, restrict, request information, review or decline first use without relying on internal history.
Actors	Customer / Risk / Credit Partner
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled

	when configured and confirmed.
Main Flow	User initiates first-time eligibility → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	First-Time Eligibility completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.
APPROVED_STARTER_LIMIT	Reusable conservative starter exposure.
APPROVED_TRANSACTION_ONLY	Only current transaction approved.
APPROVED_WITH_DOWN_PAYMENT	Higher customer contribution reduces financed exposure.
MORE_INFORMATION_REQUIRED	No decision until defined evidence supplied.
MANUAL_REVIEW	Human/partner review required.
DECLINED	No financing.
TEMPORARILY_BLOCKED	Security/fraud condition prevents decision.

CRD-005 — Limit Graduation

Field	Specification
Feature ID	CRD-005
Feature Name	Limit Graduation
Purpose	Review future limits using repayment and affordability evidence.
Actors	Risk
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates limit graduation → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit

	and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Limit Graduation completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CRD-006 — Offer Engine

Field	Specification
Feature ID	CRD-006
Feature Name	Offer Engine
Purpose	Return responsible alternatives such as higher down payment or lower financed amount.
Actors	Customer / Risk
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates offer engine → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.

Acceptance Criteria	Offer Engine completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.
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BNPL-001 — Pay-Later Purchase

Field	Specification
Feature ID	BNPL-001
Feature Name	Pay-Later Purchase
Purpose	Create an approved installment purchase.
Actors	Customer / Merchant
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates pay-later purchase → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Pay-Later Purchase completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

BNPL-002 — Installment Plans

Field	Specification
Feature ID	BNPL-002
Feature Name	Installment Plans
Purpose	Present configurable schedules and total-payable disclosure.
Actors	Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates installment plans → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only

	when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Installment Plans completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

BNPL-003 — First Payment

Field	Specification
Feature ID	BNPL-003
Feature Name	First Payment
Purpose	Collect required initial payment before activating financing/release.
Actors	Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates first payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID,

	policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	First Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

Specific acceptance: for a 2,000 ETB illustrative Pay-in-4 plan, the 500 ETB first payment must be SUCCESS before the remaining 1,500 ETB financed exposure is activated and merchant release is confirmed. PENDING/UNKNOWN remains on hold.

BNPL-004 — Flexible Down Payment

Field	Specification
Feature ID	BNPL-004
Feature Name	Flexible Down Payment
Purpose	Allow a larger upfront payment without increasing approved credit.
Actors	Customer
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates flexible down payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Flexible Down Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-001 — Payment Method Selection

Field	Specification
Feature ID	PAY-001
Feature Name	Payment Method Selection
Purpose	Choose an available, supported repayment method.
Actors	Customer

Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates payment method selection → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Payment Method Selection completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-002 — Telebirr Payment

Field	Specification
Feature ID	PAY-002
Feature Name	Telebirr Payment
Purpose	Initiate/confirm Telebirr payment using verified public/commercial capability.
Actors	Customer
Priority	MVP*
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates telebirr payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.

Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Telebirr Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

Provider note: internal adapter behavior is specified, but external endpoint/authentication/sandbox details are taken only from official Ethiopia-specific documentation or written provider onboarding. Missing details are marked **REQUIRES CONFIRMATION**.

PAY-003 — CBE Birr Payment

Field	Specification
Feature ID	PAY-003
Feature Name	CBE Birr Payment
Purpose	Support CBE Birr only through confirmed provider/gateway arrangement.
Actors	Customer
Priority	MVP*
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates cbe birr payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	CBE Birr Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

Provider note: internal adapter behavior is specified, but external endpoint/authentication/sandbox details are taken only from official Ethiopia-specific documentation or written provider onboarding. Missing details are marked **REQUIRES CONFIRMATION**.

PAY-004 — M-PESA Ethiopia Payment

Field	Specification
Feature ID	PAY-004
Feature Name	M-PESA Ethiopia Payment
Purpose	Support M-PESA Ethiopia through confirmed provider/gateway arrangement.
Actors	Customer
Priority	MVP*
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates m-pesa ethiopia payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	M-PESA Ethiopia Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

Provider note: internal adapter behavior is specified, but external endpoint/authentication/sandbox details are taken only from official Ethiopia-specific documentation or written provider onboarding. Missing details are marked **REQUIRES CONFIRMATION**.

PAY-005 — Bank Payment

Field	Specification
Feature ID	PAY-005
Feature Name	Bank Payment
Purpose	Accept supported bank payment rails with reference/reconciliation controls.
Actors	Customer / Finance
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.

Main Flow	User initiates bank payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Bank Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-006 — Payment Gateway

Field	Specification
Feature ID	PAY-006
Feature Name	Payment Gateway
Purpose	Use licensed gateway to reach multiple Ethiopian payment methods.
Actors	Customer / Platform
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates payment gateway → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.

Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Payment Gateway completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-007 — Payment Confirmation

Field	Specification
Feature ID	PAY-007
Feature Name	Payment Confirmation
Purpose	Authoritatively confirm provider outcome and post financial effects once.
Actors	Platform
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates payment confirmation → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Payment Confirmation completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-008 — Payment Retry

Field	Specification
Feature ID	PAY-008
Feature Name	Payment Retry
Purpose	Retry safely using idempotency without double charging.
Actors	Customer / Platform

Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates payment retry → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Payment Retry completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-009 — Automatic Repayment

Field	Specification
Feature ID	PAY-009
Feature Name	Automatic Repayment
Purpose	Auto-collect only with explicit mandate/provider support.
Actors	Customer / Platform
Priority	Later
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates automatic repayment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.

Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Automatic Repayment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-010 — Payment Reminders

Field	Specification
Feature ID	PAY-010
Feature Name	Payment Reminders
Purpose	Notify before/during overdue according to policy.
Actors	Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates payment reminders → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Payment Reminders completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

PAY-011 — Failed / Pending Payment

Field	Specification
Feature ID	PAY-011
Feature Name	Failed / Pending Payment
Purpose	Handle FAILED separately from PENDING/UNKNOWN.
Actors	Platform / Support
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates failed / pending payment → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Failed / Pending Payment completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

State rule: a timeout or lost callback creates PENDING/UNKNOWN and a reconciliation job; it is not converted to FAILED merely because the app did not receive an immediate success response.

COL-001 — Overdue Payments

Field	Specification
Feature ID	COL-001
Feature Name	Overdue Payments
Purpose	Apply reminders, restrictions, review and responsible collections.
Actors	Customer / Risk / Support
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates overdue payments → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review;

	customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Overdue Payments completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

COL-002 — Restructuring / Hardship

Field	Specification
Feature ID	COL-002
Feature Name	Restructuring / Hardship
Purpose	Support approved repayment arrangements/restructuring.
Actors	Customer / Support / Credit Partner
Priority	Later
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates restructuring / hardship → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.

Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Restructuring / Hardship completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

REF-001 — Refund

Field	Specification
Feature ID	REF-001
Feature Name	Refund
Purpose	Reverse eligible purchase value and adjust obligations.
Actors	Customer / Merchant / Finance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates refund → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Refund completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

REF-002 — Partial Refund

Field	Specification
Feature ID	REF-002
Feature Name	Partial Refund
Purpose	Adjust a portion without corrupting schedule or ledger.
Actors	Customer / Merchant / Finance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled

	when configured and confirmed.
Main Flow	User initiates partial refund → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Partial Refund completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

SET-001 — Merchant Settlement

Field	Specification
Feature ID	SET-001
Feature Name	Merchant Settlement
Purpose	Pay merchant according to agreed model and track status.
Actors	Merchant / Finance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates merchant settlement → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.

Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Merchant Settlement completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

FIN-001 — Fees

Field	Specification
Feature ID	FIN-001
Feature Name	Fees
Purpose	Configure lawful fees without hard-coding.
Actors	Finance / Admin
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates fees → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Fees completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

HIS-001 — Customer History

Field	Specification
Feature ID	HIS-001
Feature Name	Customer History
Purpose	Show and use verified purchase/repayment history.

Actors	Customer / Risk
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates customer history → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Customer History completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

RPT-001 — Merchant Reports

Field	Specification
Feature ID	RPT-001
Feature Name	Merchant Reports
Purpose	Provide sales, BNPL, refund, fee and settlement reporting.
Actors	Merchant
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates merchant reports → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.

User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Merchant Reports completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

RPT-002 — Customer Reports

Field	Specification
Feature ID	RPT-002
Feature Name	Customer Reports
Purpose	Provide purchases, payments, due dates and statements.
Actors	Customer
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates customer reports → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Customer Reports completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

RSK-001 — Fraud Detection

Field	Specification
Feature ID	RSK-001
Feature Name	Fraud Detection
Purpose	Detect identity, account, merchant, transaction and refund fraud.
Actors	Risk / Compliance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates fraud detection → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Fraud Detection completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

DSP-001 — Disputes

Field	Specification
Feature ID	DSP-001
Feature Name	Disputes
Purpose	Investigate financial and merchant disputes with evidence.
Actors	Customer / Merchant / Support
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates disputes → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without

	traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Disputes completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

SUP-001 — Customer Support

Field	Specification
Feature ID	SUP-001
Feature Name	Customer Support
Purpose	Track assistance, attachments, escalation and resolution.
Actors	Customer / Support
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates customer support → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.

Acceptance Criteria	Customer Support completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.
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ADM-001 — Admin Platform

Field	Specification
Feature ID	ADM-001
Feature Name	Admin Platform
Purpose	Operate customers, merchants, finance, risk, support and configuration.
Actors	Admin Roles
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates admin platform → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Admin Platform completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

IAM-001 — Roles & Permissions

Field	Specification
Feature ID	IAM-001
Feature Name	Roles & Permissions
Purpose	Apply least privilege and separation of duties.
Actors	All privileged roles
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates roles & permissions → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only

	when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Roles & Permissions completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

AUD-001 — Audit Logs

Field	Specification
Feature ID	AUD-001
Feature Name	Audit Logs
Purpose	Record significant financial/admin state changes immutably.
Actors	Compliance / Audit
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates audit logs → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID,

	policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Audit Logs completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

ENT-001 — Enterprise Merchant Connection

Field	Specification
Feature ID	ENT-001
Feature Name	Enterprise Merchant Connection
Purpose	Connect approved enterprise POS/e-commerce systems.
Actors	Enterprise Merchant
Priority	Should
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates enterprise merchant connection → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Enterprise Merchant Connection completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

RPT-003 — Management Reporting

Field	Specification
Feature ID	RPT-003
Feature Name	Management Reporting
Purpose	Track GMV, repayment, overdue, fraud, revenue and operational health.
Actors	Management
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.

Main Flow	User initiates management reporting → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Management Reporting completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CFG-001 — System Configuration

Field	Specification
Feature ID	CFG-001
Feature Name	System Configuration
Purpose	Version and approve plans, limits, fees, provider settings and policies.
Actors	Authorized Admin
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates system configuration → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.

Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	System Configuration completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CMP-001 — Terms & Consent

Field	Specification
Feature ID	CMP-001
Feature Name	Terms & Consent
Purpose	Record exact terms/offer/privacy consent versions.
Actors	Customer / Compliance
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates terms & consent → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Terms & Consent completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

CMP-002 — Complaint Management

Field	Specification
Feature ID	CMP-002
Feature Name	Complaint Management
Purpose	Manage complaints, SLA evidence and regulatory escalation.
Actors	Customer / Support / Compliance

Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates complaint management → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Complaint Management completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

REC-001 — Reconciliation

Field	Specification
Feature ID	REC-001
Feature Name	Reconciliation
Purpose	Compare provider, bank, settlement and internal ledger records.
Actors	Finance / Operations
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates reconciliation → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending

	states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Reconciliation completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

SEC-001 — Device & Session Security

Field	Specification
Feature ID	SEC-001
Feature Name	Device & Session Security
Purpose	Protect sessions, devices, tokens and high-risk actions.
Actors	All users
Priority	MVP
Preconditions	Authenticated/identified actor as applicable; required upstream verification complete; referenced records active; provider-dependent actions only enabled when configured and confirmed.
Main Flow	User initiates device & session security → system validates inputs and permissions → applies business/risk rules → records state change → returns clear result → sends required notifications.
Alternative Flows	Eligible alternative payment/identity/merchant route; manual review; customer cancels before irreversible financial action; provider fallback only when approved.
Failure Flows	Validation failure; unauthorized request; duplicate request; dependency unavailable; PENDING/UNKNOWN external state; fraud/risk hold; configuration disabled.
Business Rules	Rules are configuration/version controlled; no financial state changes without traceable source and approval; provider-specific rules remain isolated; credit and payment decisions are not inferred from UI success.
Required Information	Actor ID; transaction/context IDs; timestamps; versioned policy/configuration; feature-specific data; provider reference where external.
Validation	Schema + business validation; ownership/permission; amount/currency; state transition; duplicate/idempotency checks; risk rules where applicable.
User Interface	Clear current state, next action, amount/date where relevant, loading/pending states, accessible error and support path.
Notifications	Event-driven push/SMS/email according to user preference, severity and regulatory/business requirement.
Security	Least privilege; encrypted transport/storage; rate limits; step-up authentication for sensitive actions; sensitive-data masking; anti-replay/idempotency.
Financial Impact	If money-related: use integer minor units or exact decimal; create balanced ledger postings; never mutate a balance as sole source of truth; financial effect only after authoritative status.
Audit Requirements	Record actor, action, old/new state, reason, device/source, correlation ID, policy/config version and timestamp.
Data Requirements	Normalized relational records, foreign keys, state constraints, indexed lookup/reference fields, immutable history for critical events.
Dependencies	Identity/RBAC, configuration, audit; plus feature-specific payment/KYC/risk/notification provider dependencies.
Acceptance Criteria	Device & Session Security completes its intended business outcome; invalid/duplicate/unauthorized requests are rejected safely; failures do not corrupt financial or credit state; relevant UI, audit and notifications match the final state.

4. Traceability

Feature ID	Related Business Requirements
AUTH-001	See BRD/SRS cross-cutting requirements
AUTH-002	See BRD/SRS cross-cutting requirements
AUTH-003	See BRD/SRS cross-cutting requirements
CUS-001	See BRD/SRS cross-cutting requirements
KYC-001	See BRD/SRS cross-cutting requirements
KYC-002	See BRD/SRS cross-cutting requirements
MER-001	BR-007, BR-008, BR-009, BR-021, BR-023
MER-002	BR-007, BR-008, BR-009, BR-021, BR-023
MER-003	BR-007, BR-008, BR-009, BR-021, BR-023
MER-004	BR-007, BR-008, BR-009, BR-021, BR-023
MER-005	BR-007, BR-008, BR-009, BR-021, BR-023
MER-006	BR-007, BR-008, BR-009, BR-021, BR-023
QR-001	See BRD/SRS cross-cutting requirements
QR-002	See BRD/SRS cross-cutting requirements
DISC-001	See BRD/SRS cross-cutting requirements
CRD-001	BR-002, BR-003, BR-030, BR-034, BR-037
CRD-002	BR-002, BR-003, BR-030, BR-034, BR-037
CRD-003	BR-002, BR-003, BR-030, BR-034, BR-037
CRD-004	BR-002, BR-003, BR-030, BR-034, BR-037
CRD-005	BR-002, BR-003, BR-030, BR-034, BR-037
CRD-006	BR-002, BR-003, BR-030, BR-034, BR-037
BNPL-001	See BRD/SRS cross-cutting requirements
BNPL-002	See BRD/SRS cross-cutting requirements
BNPL-003	See BRD/SRS cross-cutting requirements
BNPL-004	See BRD/SRS cross-cutting requirements
PAY-001	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-002	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-003	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-004	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-005	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-006	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-007	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-008	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-009	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-010	BR-005, BR-007, BR-012, BR-014, BR-015
PAY-011	BR-005, BR-007, BR-012, BR-014, BR-015
COL-001	See BRD/SRS cross-cutting requirements
COL-002	See BRD/SRS cross-cutting requirements
REF-001	See BRD/SRS cross-cutting requirements
REF-002	See BRD/SRS cross-cutting requirements
SET-001	See BRD/SRS cross-cutting requirements
FIN-001	See BRD/SRS cross-cutting requirements
HIS-001	See BRD/SRS cross-cutting requirements
RPT-001	See BRD/SRS cross-cutting requirements
RPT-002	See BRD/SRS cross-cutting requirements
RSK-001	See BRD/SRS cross-cutting requirements
DSP-001	See BRD/SRS cross-cutting requirements
SUP-001	See BRD/SRS cross-cutting requirements
ADM-001	See BRD/SRS cross-cutting requirements
IAM-001	See BRD/SRS cross-cutting requirements
AUD-001	See BRD/SRS cross-cutting requirements
ENT-001	See BRD/SRS cross-cutting requirements
RPT-003	See BRD/SRS cross-cutting requirements
CFG-001	See BRD/SRS cross-cutting requirements
CMP-001	See BRD/SRS cross-cutting requirements
CMP-002	See BRD/SRS cross-cutting requirements
REC-001	See BRD/SRS cross-cutting requirements
SEC-001	See BRD/SRS cross-cutting requirements

References

- **[ETH-NBE-01] VERIFIED — National Bank of Ethiopia.** Payment Instrument Issuers / System Operators.
<https://nbe.gov.et/payment-instrument-issuers-system-operators/>
Supports: Current licensed/commercialized payment instrument issuers and payment system operators, including Telebirr, Kacha, M-PESA Ethiopia, EthSwitch, ArifPay, Chapa and SantimPay.
- **[ETH-NBE-02] VERIFIED — National Bank of Ethiopia.** Financial Stability Report — March 2026.
https://nbe.gov.et/wp-content/uploads/2026/03/Financial-Stability-Report_MARCH_2026.pdf
Supports: Current payment-sector regulatory framework, digital-finance context and systemic risk context.
- **[ETH-NBE-03] VERIFIED — National Bank of Ethiopia.** National Digital Payments Strategy 2026–2030 — Draft.
https://nbe.gov.et/wp-content/uploads/2025/12/Ethiopia_NDPS_Draft_F.pdf
Supports: Interoperability direction and digital-payment strategy context; document is a draft.
- **[ETH-NBE-04] VERIFIED — National Bank of Ethiopia.** Financial Consumer Protection — Complaint Handling.
<https://nbe.gov.et/fcpe/complaint/>
Supports: Complaint escalation after first contacting the financial service provider; NBE review where unresolved/no response within 10 business days.
- **[ETH-DP-01] VERIFIED — Federal Democratic Republic of Ethiopia / Ministry of Justice.** Personal Data Protection Proclamation No. 1321/2024.
<https://justice.gov.et/en/law/personal-data-protection-proclamation/>
Supports: Ethiopian personal-data protection framework.
- **[ETH-QR-01] VERIFIED — EthSwitch.** Interoperable QR Standard.
<https://ethswitch.com/wp-content/uploads/2024/12/Interoperable-QR-Standard.pdf>
Supports: Interoperable QR design direction for merchant payments and limitations of closed-loop QR.
- **[TEL-01] VERIFIED — Ethio telecom.** Telebirr Developer Portal — Getting Started.
<https://developer.ethiotelcom.et/docs/GettingStarted>
Supports: Publicly documented Telebirr business integration categories including C2B, B2B, H5, Mini App and subscription payment.
- **[TEL-02] VERIFIED — Ethio telecom.** Telebirr H5 C2B Web Payment Integration.
<https://developer.ethiotelcom.et/docs/category/h5-c2b-web-payment-integration>
Supports: Public create-order, checkout, query/status, callback and refund integration documentation.
- **[TEL-03] VERIFIED — Ethio telecom.** Telebirr Schedule Payment / CreateDisburseOrder.
<https://developer.ethiotelcom.et/docs/Schedule%20Payment/CreateDisburseOrder>
Supports: Public documentation for mandate-based scheduled deduction; BNPL repayment use requires commercial/regulatory confirmation.
- **[CBE-01] VERIFIED — Commercial Bank of Ethiopia.** CBE Birr.
<https://combanketh.et/ways-of-banking/cbe-birr>
Supports: Official description of CBE Birr customer/mobile-money services.
- **[CBE-02] VERIFIED — Commercial Bank of Ethiopia.** CBE Products and Services.
https://combanketh.et/cbeapi/uploads/CBE_Products_and_services_English_84e511c740.pdf
Supports: CBE Birr supports transfers, cash out, airtime, goods purchase and bill payment; no equivalent public direct BNPL integration specification verified.
- **[MPE-01] VERIFIED — Safaricom Ethiopia.** M-PESA Ethiopia Merchant Terms / Business Use.
<https://www.safaricom.et/>
Supports: M-PESA Ethiopia supports merchant payments/business arrangements; exact BNPL API onboarding and sandbox require provider confirmation.

- **[QCB-01] VERIFIED — Qatar Central Bank.** Buy Now Pay Later Regulation.
<https://www.qcb.gov.qa/Services/Financial%20Technology/QCB%20BNPL%20Regulation.pdf>
Supports: Qatar definition, eligibility and regulated BNPL-provider framework. Qatar law is not Ethiopian law.
- **[QTR-TC] VERIFIED — PayLater Qatar.** PayLater General Terms and Conditions — client-provided 20-page copy.
Client-provided PDF reviewed 21 August 2026
Supports: Customer/merchant/PayLater roles, eligibility, 25% first payment, PayMore, limits, repayment, default, restructuring, disputes, data handling and account rules.
- **[QTR-PL-01] VERIFIED — PayLater Qatar.** PayLater Terms and Conditions.
<https://paylaterapp.com/terms-and-conditions/>
Supports: Current public PayLater Qatar contractual product model.
- **[QTR-PL-02] VERIFIED — PayLater Qatar.** How can I sign up for PayLater?.
<https://help.paylaterapp.com/en/support/solutions/articles/154000139292-how-can-i-sign-up-for-paylater->
Supports: Phone OTP, QID capture and liveness-based customer onboarding.
- **[QTR-PL-03] VERIFIED — PayLater Qatar.** What is my credit limit and how much can I spend?.
<https://help.paylaterapp.com/en/support/solutions/articles/154000139303-what-is-my-credit-limit-and-how-much-can-i-spend-with-paylater->
Supports: Credit limit uses creditworthiness, PayLater repayment history and account usage.
- **[QTR-PL-04] VERIFIED — PayLater Qatar.** Where can I use PayLater?.
<https://help.paylaterapp.com/en/support/solutions/articles/154000139291-where-can-i-use-paylater->
Supports: Use across online and in-store merchants.
- **[QTR-CBQ-01] VERIFIED — Commercial Bank Qatar / Vodafone Qatar.** Vodafone Easy-Buy / CBQ Buy Now Pay Later.
<https://www.vodafone.qa/en/buynowpaylater>
Supports: Bank-card installment model for online and selected physical stores; full purchase reduces card limit and customer repays equal monthly installments.
- **[QTR-VF-01] VERIFIED — Vodafone Qatar.** Vodafone Partners with Commercial Bank to Offer BNPL.
<https://www.vodafone.qa/en/about-us/media/press-release/vodafone-partners-with-commercial-bank-to-offer-buy-now-pay-later-installment-plan>
Supports: Qatar device-installment model through bank partnership and merchant stores.